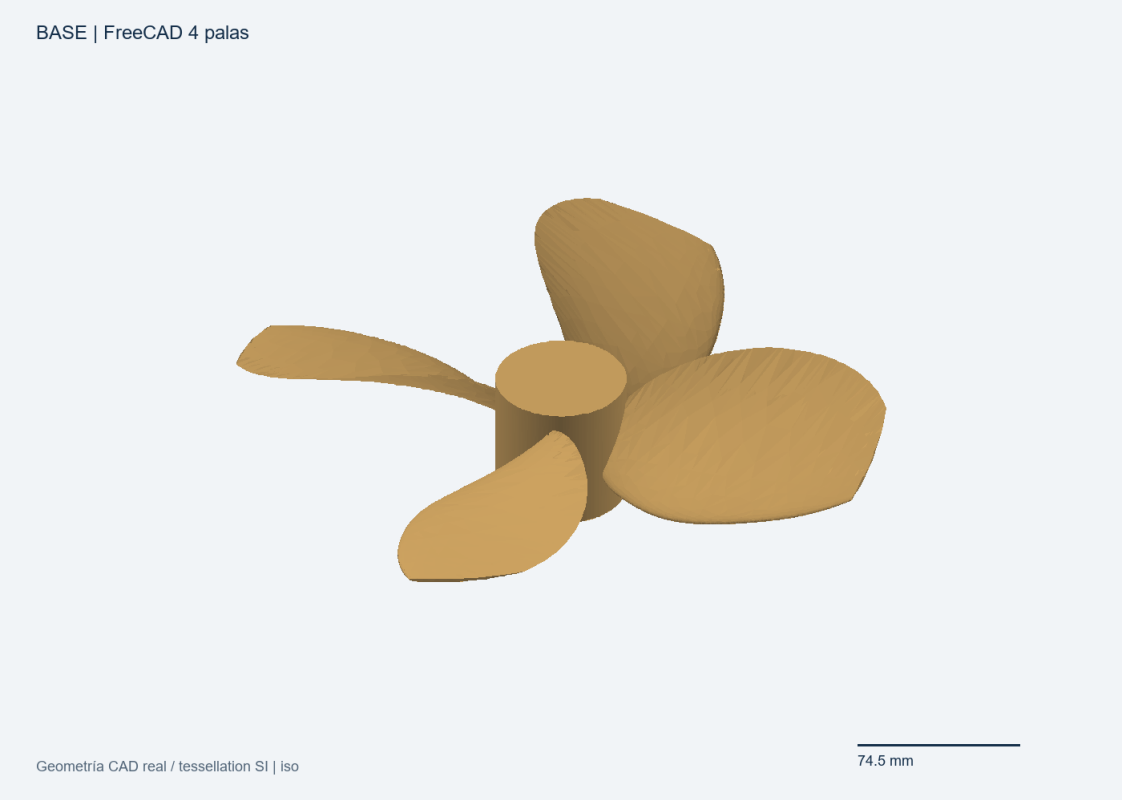
CantaX
Propeller Lab

Parametric Marine Propeller Engineering

V1.0 publication report | 12 October 2026



A repeatable Excel-to-CAD-to-calculation-to-report workflow, developed with AI assistance and documented numerical checks. This report uses real saved synthetic engineering outputs. Author review and physical validation remain necessary before engineering decisions.

Integration verified	Preliminary models	Not implemented/validated
Native CAD and file reopen; Excel input/recalculation; native solver execution	Analytical BEM, shaft-control scenario, BASE linear elasticity	Validated 3D CFD, cavitation, fatigue, corrosion, allowable stress and service certification

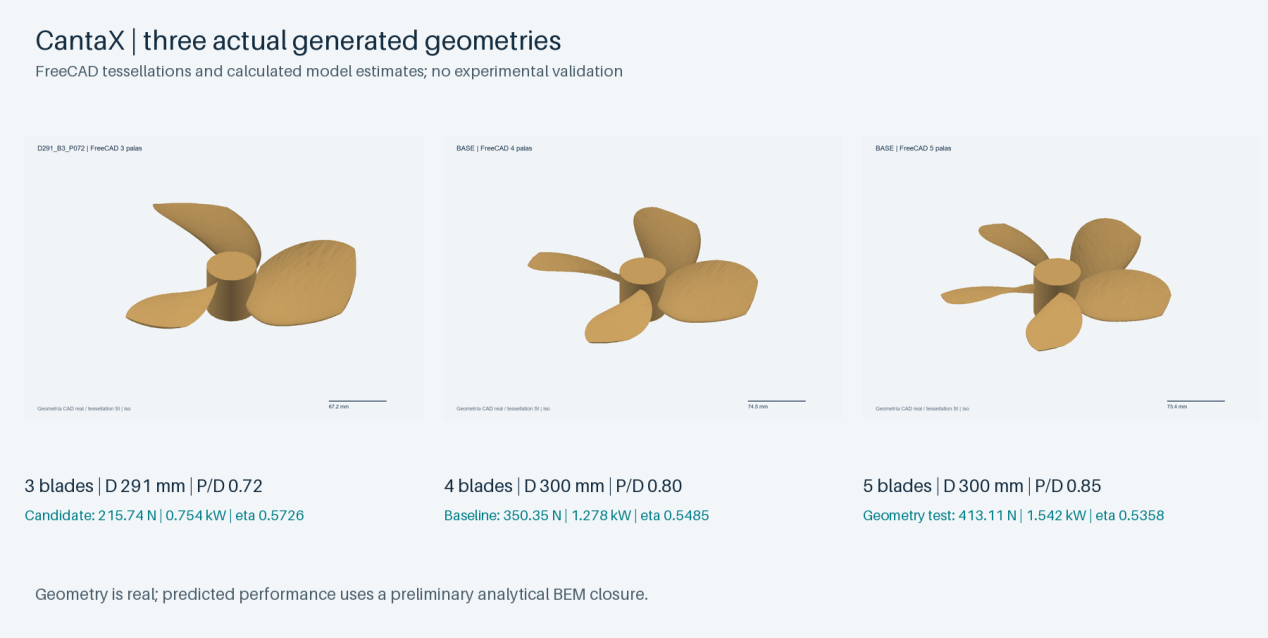
Independent use requires the documented local programs and licenses. No vendor application, installer, runtime or font file is distributed.

01 | Calculated results and input changes

All three cases use D=0.300 m, Va=2 m/s, seawater density 1025 kg/m3 and reference bronze density 7640 kg/m3. Figures are model estimates; decimal precision permits reproducibility checks, not a claim of physical accuracy.

Case	B	P/D	RPM	T N	Q Nm	P W	eta	kg
baseline	4	0.80	900	350.35	13.555	1277.56	0.5485	2.7233
rpm_1000	4	0.80	1000	480.86	18.021	1887.19	0.5096	2.7233
five_blades	5	0.85	900	413.11	16.361	1542.02	0.5358	2.4387

RPM change: the geometry signature and CAD volume remain identical at 0.0003564570746956503 m3. Geometry change: B=5 and P/D=0.85 at 900 RPM gives 0.000319203190234742 m3. CAD, STEP, mesh and numerical artifact hashes differ; mass/volume decrease by 10.4512%.



At fixed nominal expanded area ratio, increasing blade count reduces individual chord. A lower mass does not establish a stronger or more efficient propeller.

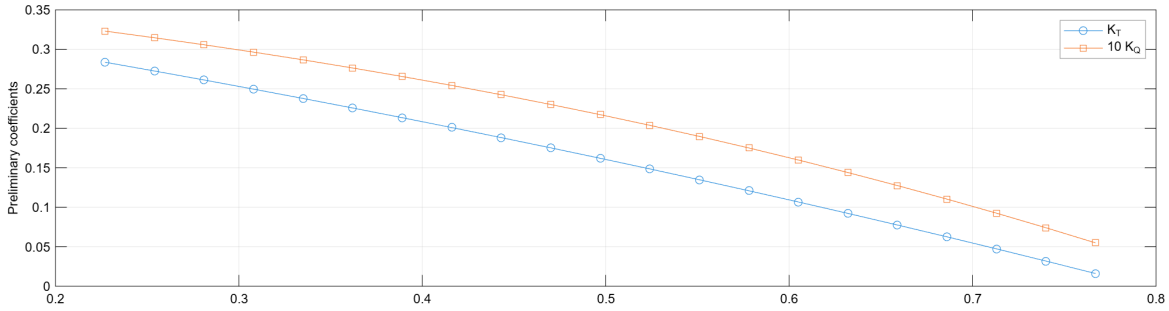
02 | Analytical BEM and physical plausibility

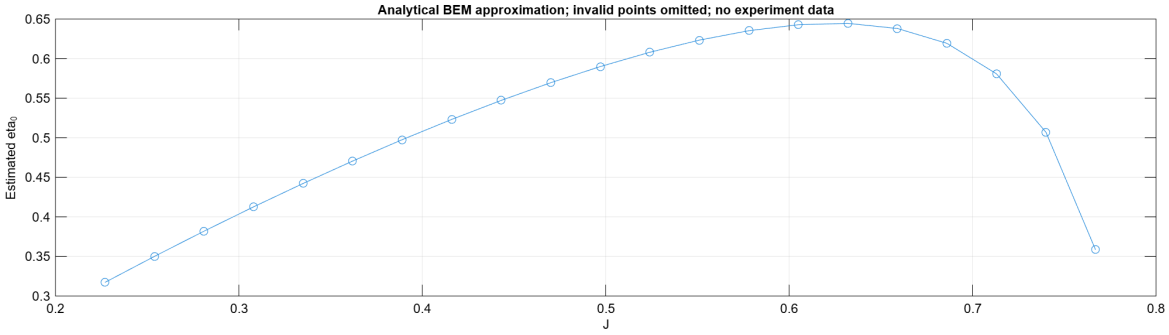
The axial blade-element/momentum model solves annular induced velocities with Prandtl tip/hub corrections. Symmetric thin-airfoil lift Cl=2*pi*alpha and Cd=2*Cf from ITTC-1957 friction are explicit analytical closures. No measured section polars, pressure/separation drag, transition, roughness, wake, cavitation or free-surface validation are supplied.

Definition	Equation / baseline value
Rotation	n=RPM/60; omega=2*pi*n
Advance ratio	J=Va/(n*D) = 0.444444444
Thrust coefficient	KT=T/(rho*n^2*D^4) = 0.187546714
Torque coefficient	KQ=Q/(rho*n^2*D^5) = 0.024187924
Power / efficiency	P=omega*Q; eta=T*Va/P = 0.548464440

An optimistic full ideal actuator disk gives induced velocity 0.848720617 m/s and minimum power 998.046 W for the stated thrust. The estimate 1277.563 W exceeds this lower bound; its efficiency is below the ideal 0.70207. This elementary coherence check does not calibrate the propeller model.

Radial refinement from 60 to 120 annuli changes T by 0.02384% and Q by 0.02732%. Guarded positive-thrust outputs are withheld outside the declared domain. Numerical convergence of these equations is distinct from physical validation.





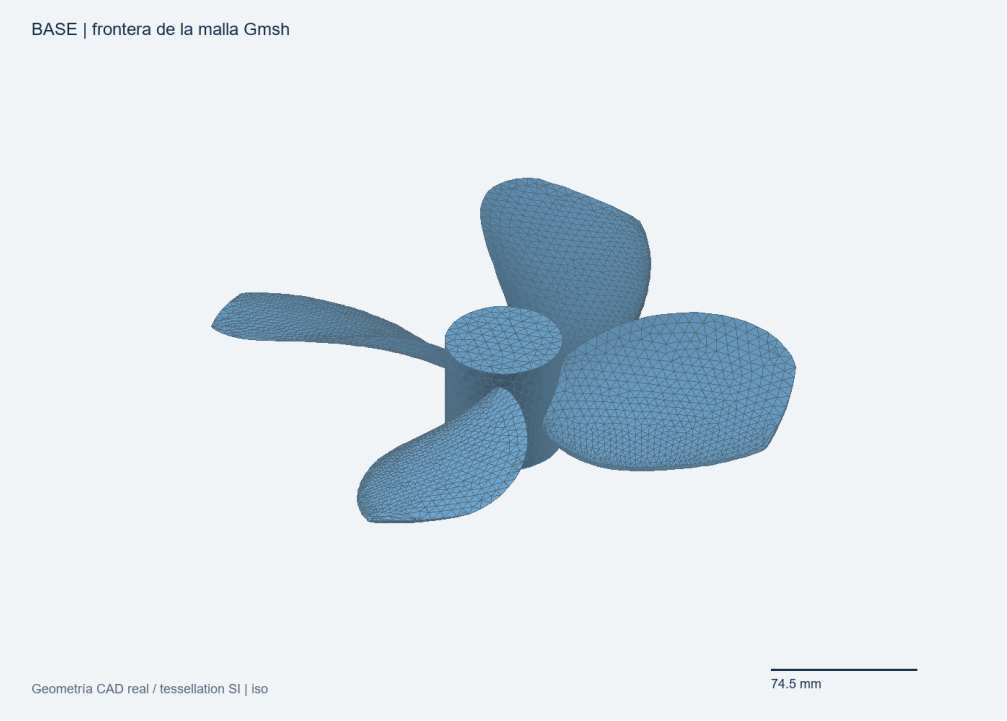
03 | CAD, geometry and mesh evidence

FreeCAD Python creates original section curves, radial lofts, rotated blades and a fused hub, then clips the tip envelope. Geometry is parameterized by the input/scripts. The saved FCStd examples are editable generated solids, not a native feature-tree configurator. FCStd and STEP are reopened and checked for validity, closedness, solid count, bounding box and volume.

BASE quantity	Calculated value
Closed solid	Valid / closed / one solid
Volume	3.56e-04 m3
Reference mass	2.723332 kg
Nominal EAR	0.550
Clipped strip EAR estimate	0.540214
Axial CAD inertia	0.0130103 kg m2

CAD uses millimetres internally; published lengths/areas/volumes convert to m/m2/m3. BEM uses nominal radial strips rather than the exact clipped 3D blade. Skew, rake, thickness and tip clipping are geometrically present but not resolved as 3D flow effects.

Mesh	Tetrahedra	Volume error	Inverted / degenerate	q < 0.05
1	8996	0.930%	0 / 0	3
2	19055	0.485%	0 / 0	3



04 | Preliminary linear elastic analysis

CalculiX 2.22 first solves a synthetic uniaxial cube benchmark: 100 N, expected stress 10000 Pa, displacement 5e-9 m and reaction -100 N. The solver output agrees within the stated tolerance. This verifies one analytic elastic problem, not a propeller design.

For BASE, Gmsh tetrahedra feed C3D4 linear isotropic elasticity with an ideal rigid hub constraint. Annular BEM thrust/torque become conservative equivalent nodal loads, with centrifugal loading from mesh mass and RPM. The distribution is not a computed suction/pressure field.

BASE elastic result	Value / interpretation
Material	NAB C95800: public typical density 7640 kg/m3, E=113.7635 GPa, nu=0.32
Maximum displacement	35.421 micrometres
Integration-point von Mises	16.929 MPa
Relative force residual	2.02e-04
Relative moment residual	2.81e-06
Strength/fatigue/corrosion/service	NOT CALCULATED; no allowable stress or certified safety factor

Three low-quality slivers per BASE mesh remain recorded. Two mesh sizes demonstrate geometric volume variation only; they do not establish convergence of displacement or stress. Hub fillets, real supports, manufacturing route, fatigue loads and admissibles are required before structural approval.

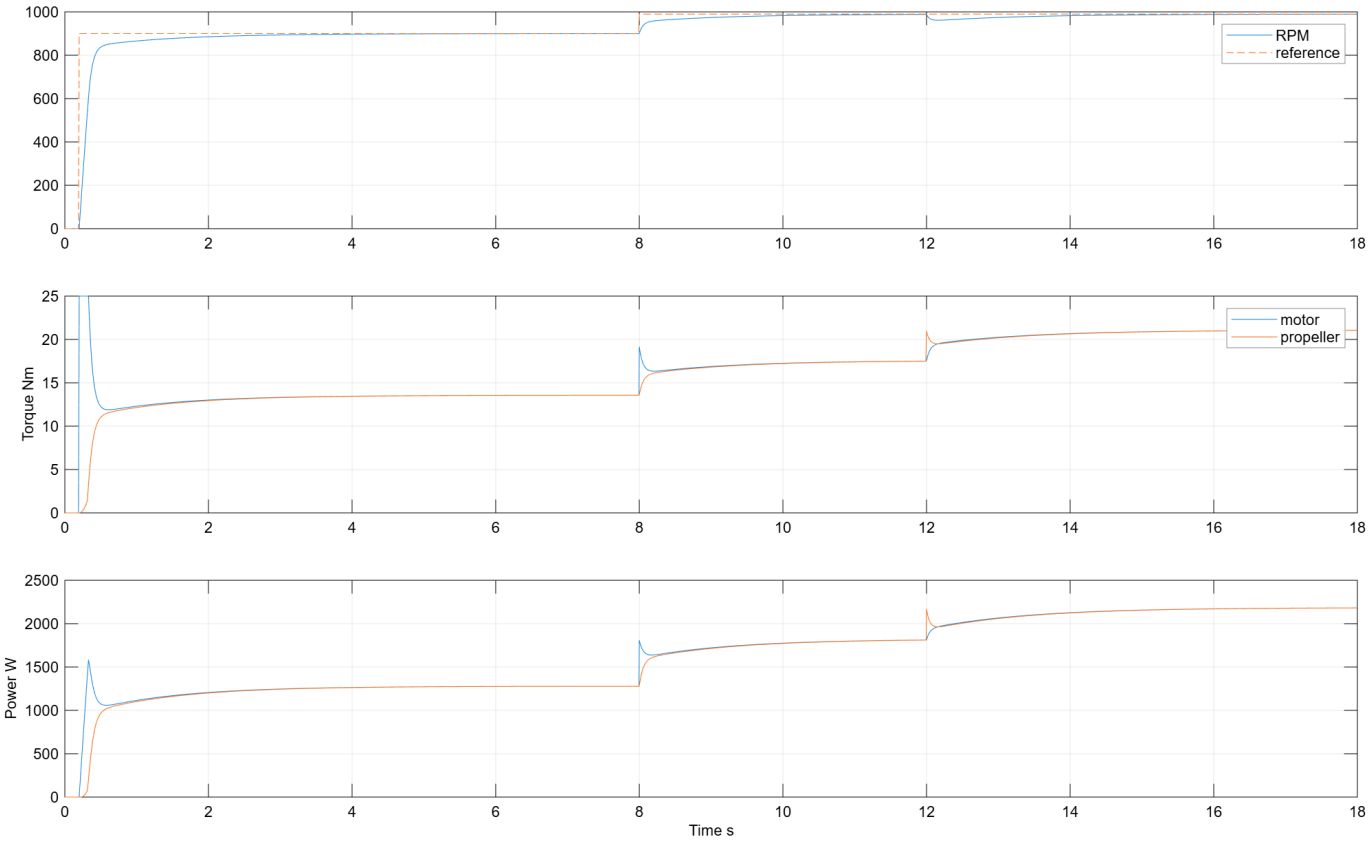
Material properties are a small set of attributed reference facts, not supplier certification or a redistributed material database. 316L and 6061 options retain absent constitutive/service properties as NOT CALCULATED.

05 | Simulink scenario and motor limits

An actual Simulink SLX stores its model workspace, steady BEM torque lookup and scenario parameters. It simulates startup, a +10% RPM reference step and a +20% load step. Saving, reopening and resimulating gives maximum trace difference zero. Inertia, PI gains, torque cap and extra load are explicit user scenarios.

Case	Final ref RPM	Final RPM	Requested Nm	Motor feasible	Energy error
baseline	990.0	989.58	21.076	True	0.1072%
rpm_1000	1100.0	1057.54	27.605	False	0.0768%
five_blades	990.0	987.21	25.182	False	0.0655%

The 1000-RPM case requests 27.605 Nm against a 25-Nm motor cap after the load step, ending at 1057.54 RPM instead of 1100. The five-blade case also exceeds steady motor capacity. The software preserves these unfavorable outcomes rather than forcing tracking to pass.



Below the accepted positive-thrust branch, startup uses an explicit Q proportional to omega-squared surrogate. It does not model windmilling, dynamic inflow, motor electrics, gearbox dynamics, ventilation or torsional response.

06 | Thirteen-configuration screening

The study contains BASE plus 12 combinations: D=0.291/0.309 m, B=3/4/5 and P/D=0.72/0.88. All use the same reference operating inputs and EAR=0.55. Each row has its own CAD and calculated BEM results.

ID	D m	B	P/D	T N	P W	eta	kg	Rank
D291_B3_P072	0.291	3	0.72	215.7	753.5	0.5726	2.907	1
D291_B4_P072	0.291	4	0.72	223.9	783.5	0.5716	2.486	2
D291_B5_P072	0.291	5	0.72	228.8	803.2	0.5696	2.226	3
D309_B3_P072	0.309	3	0.72	300.5	1082.1	0.5553	3.481	4
D309_B4_P072	0.309	4	0.72	311.8	1123.7	0.5549	2.976	5
D309_B5_P072	0.309	5	0.72	318.4	1150.9	0.5534	2.665	6
BASE	0.300	4	0.80	350.3	1277.6	0.5485	2.723	7
D291_B5_P088	0.291	5	0.88	383.6	1423.1	0.5391	2.226	8
D291_B4_P088	0.291	4	0.88	374.4	1390.2	0.5386	2.485	9
D291_B3_P088	0.291	3	0.88	359.1	1337.9	0.5368	2.908	10
D309_B5_P088	0.309	5	0.88	514.9	1992.0	0.5170	2.665	11
D309_B4_P088	0.309	4	0.88	502.7	1947.2	0.5163	2.976	12
D309_B3_P088	0.309	3	0.88	482.3	1875.8	0.5143	3.482	13

The ranking sorts efficiency for candidates satisfying the declared thrust/power/mass/CAD guards. D291_B3_P072 is first in this sample: 215.74 N, 753.54 W, eta=0.572596 and mass=2.907484 kg. Its elastic analysis is NOT CALCULATED.

Cavitation, allowable stress, fatigue, corrosion, manufacturing and vessel integration are outside the ranking. Neither global optimality nor a recommendation for fabrication is claimed.

07 | Reproducibility and release boundaries

The release includes original source, configurable application adapters, selected sanitized JSON/CSV results, generated CAD examples, real figures and this report. Raw private logs, source-machine paths, tokens, proprietary application binaries, runtimes, installers, font files and third-party publications are excluded.

Verification	What it establishes
12 Python unit tests	Definitions/input guards, workbook structure and process/environment safety
Saved demonstration	Four real historical cases; 260 numerical identities; 52 historical solid records
Public-distribution native run	Fresh Excel input -> FreeCAD -> MATLAB/Simulink -> Gmsh/CalculiX -> Octave/Scilab -> native Excel -> PDF
PDF generation	ReportLab standard fonts and page tracking; pypdf removed; optional external Poppler rendering
Human engineering review	Required before decisions; numerical/visual checks do not certify performance

The full workflow requires Windows, Excel and MATLAB/Simulink licenses plus installed FreeCAD, Gmsh, CalculiX, Octave and Scilab. The saved-example demonstration requires only the Python dependencies. No AI API is used at runtime. MATLAB may contact its licensing service.

Primary model/material references: ITTC open-water/friction procedures; thin-airfoil theory; Prandtl BEM corrections; Copper Development Association C95800; Outokumpu 316L; Hydro 6061. Full source links and limits are in docs/BEM.md, docs/STRUCTURAL_ANALYSIS.md and config/hydro_references.json. No upstream PDF or vendor documentation is included.

Next steps: section polars and open-water experiments; CFD/cavitation validation; converged structural models and real admissibles; uncertainty-aware constrained selection; independent packaging and license review. V1.0 remains a preliminary engineering demonstrator.

Repository: aaronexposito0-creator/cantax-propeller-lab